

Question: Why is dynamic pipelining more complex than static/traditional pipelining?

Answer: Dynamic pipelining is more complex than static pipelining for two main reasons:

1. **Branch Prediction:** It uses branch prediction, which requires the commit unit to discard all results in the execution units if a branch is mispredicted. [PDF](#)
2. **Superscalar Execution:** It is typically combined with superscalar execution, meaning each unit must handle issuing or committing multiple (four to six) instructions every clock cycle.

[PDF](#)

Question: Briefly explain superscalar and superpipelining.

Answer: * **Superpipelining:** It simply means using longer pipelines, often with eight or more stages, to increase a microprocessor's speed. [PDF](#)

- **Superscalar:** It involves duplicating the computer's internal components so that it can launch multiple instructions in every single pipeline stage. [PDF](#)